MeshX platform for banks

Scope and screens. What runs in your network, and what is built.

Purpose of this session

Two questions. Can it run in your network? What is ready today?

- The platform runs in your own network. No meshX control plane is needed.
- You get a capability map: what we supply, and what you supply.
- You get a clear status: built, partly built, or not started.
- You get next steps, with owners.

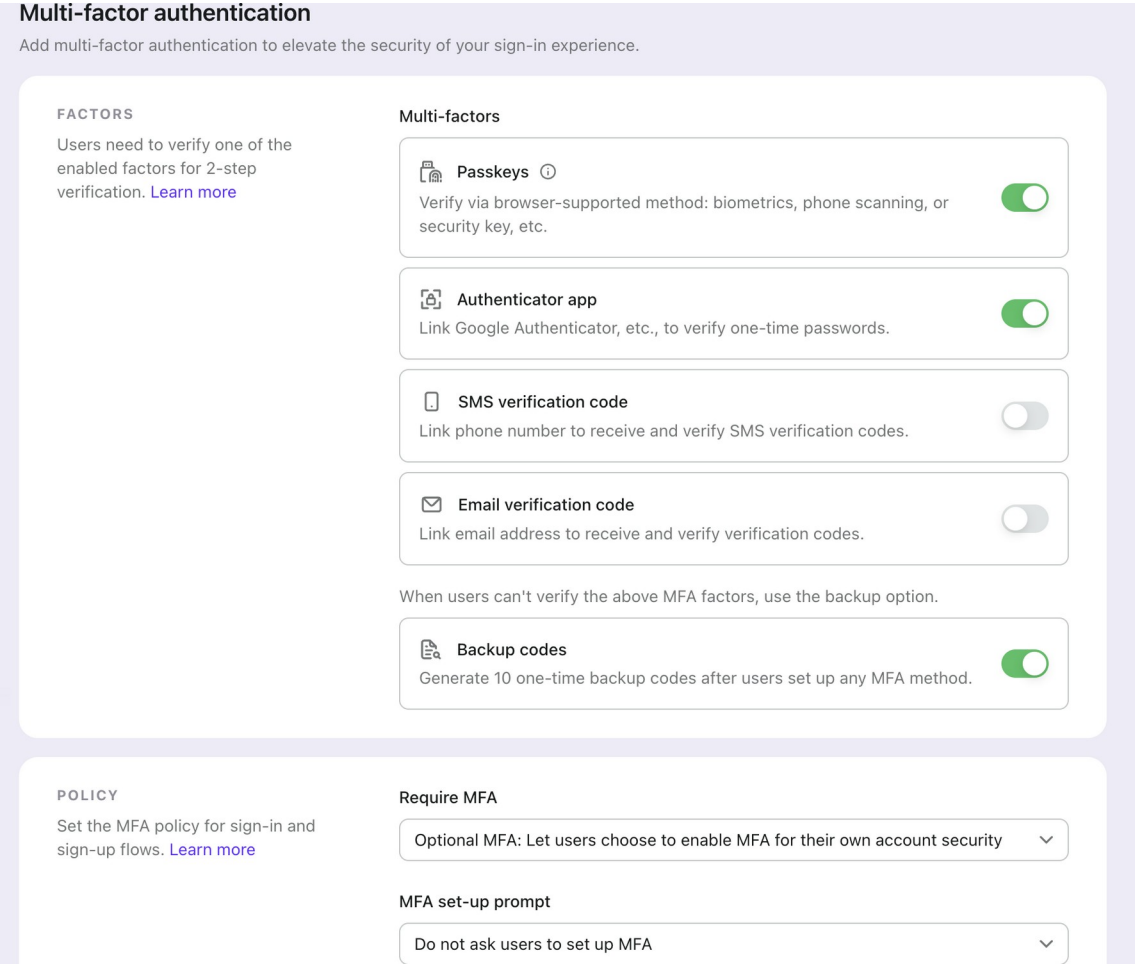
Capability map

Status of each capability, from the code today

Capability	Status	What exists
Stack operator (pull-based)	**Proven**	Kubernetes operator. Direct (Helm) or Flux mode. Ordered deploys.
Helm chart set	**Proven**	23 charts, including Postgres, Redis, Kafka, NATS, Trino and object storage
Air-gap bundle import	**Partial**	Upload and import. Checksums. In-cluster registry. Image references not yet rewritten.
Identity (Logto)	**Partial**	Logto chart and OAuth link. Product templates still default to Auth0.
LLM access	**Partial**	LiteLLM gateway. Default is the public API. Bedrock not yet tested.
Cost visibility	**Proven**	Price tables and cost per call. Visibility only.
Model routing	**Gap**	Host mapping only. No model selection.
GPU and on-prem models	**Gap**	No GPU requests. No path to an on-prem model.
Federation across entities	**Gap**	One namespace per tenant. No federation code.
Operator access	**Gap**	Tailscale only. Bastion and break-glass not defined.

Multi-factor authentication

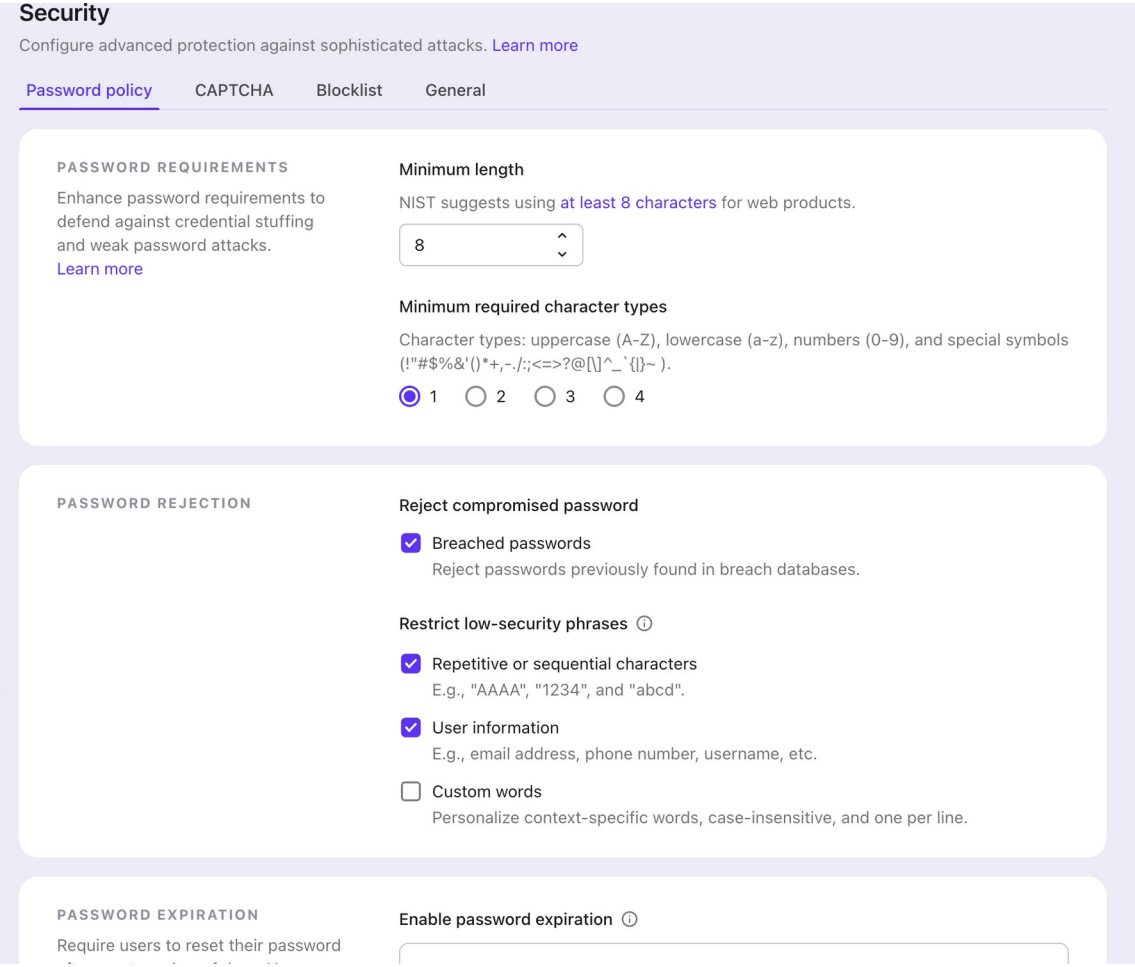
Set MFA for each tenant



- Set when MFA is required: optional or enforced.
- Select the factors the bank accepts.
- Backup codes give a recovery path. The policy stays strong.

Security settings

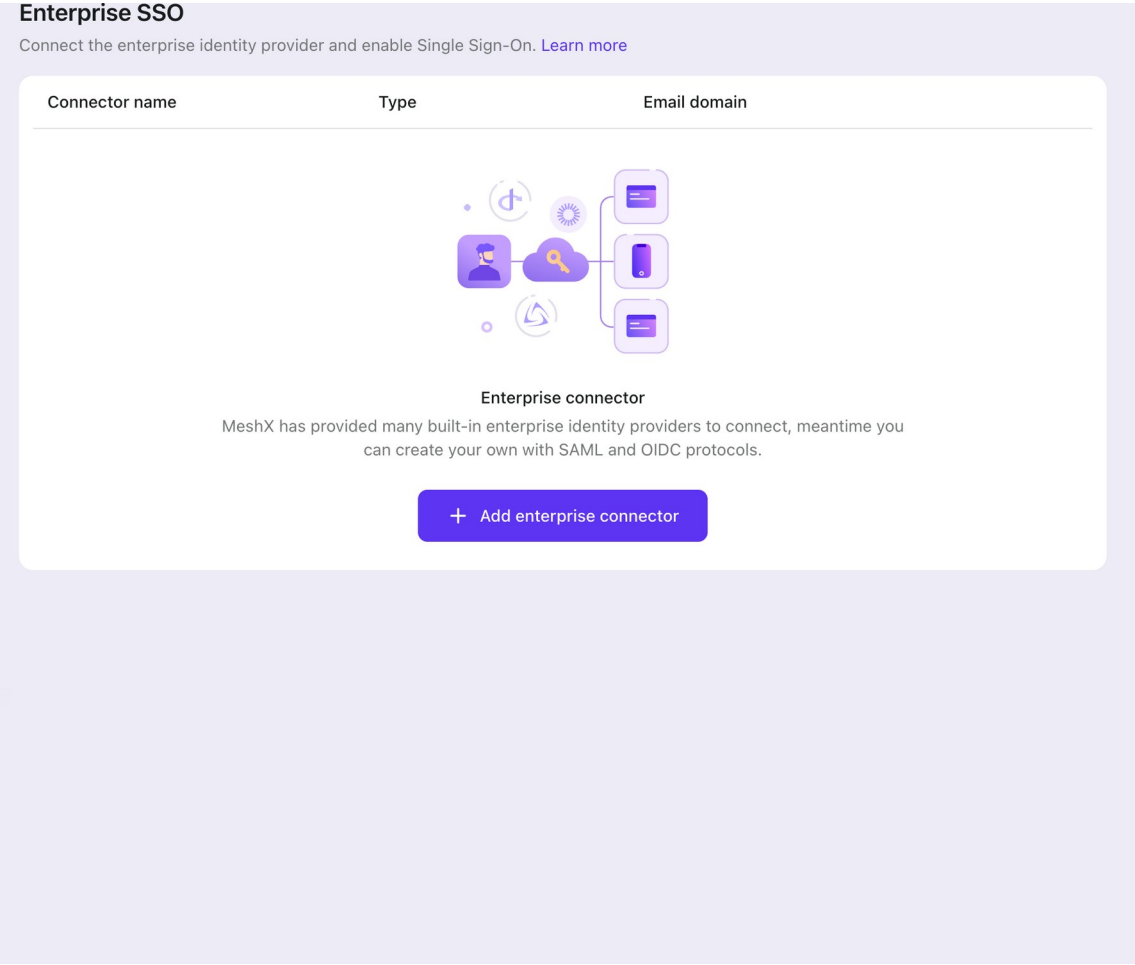
Set password, CAPTCHA and sign-up rules



- Set password rules for all users.
- Turn on CAPTCHA for sign-in and sign-up.
- Every change applies to the whole tenant.

Enterprise single sign-on

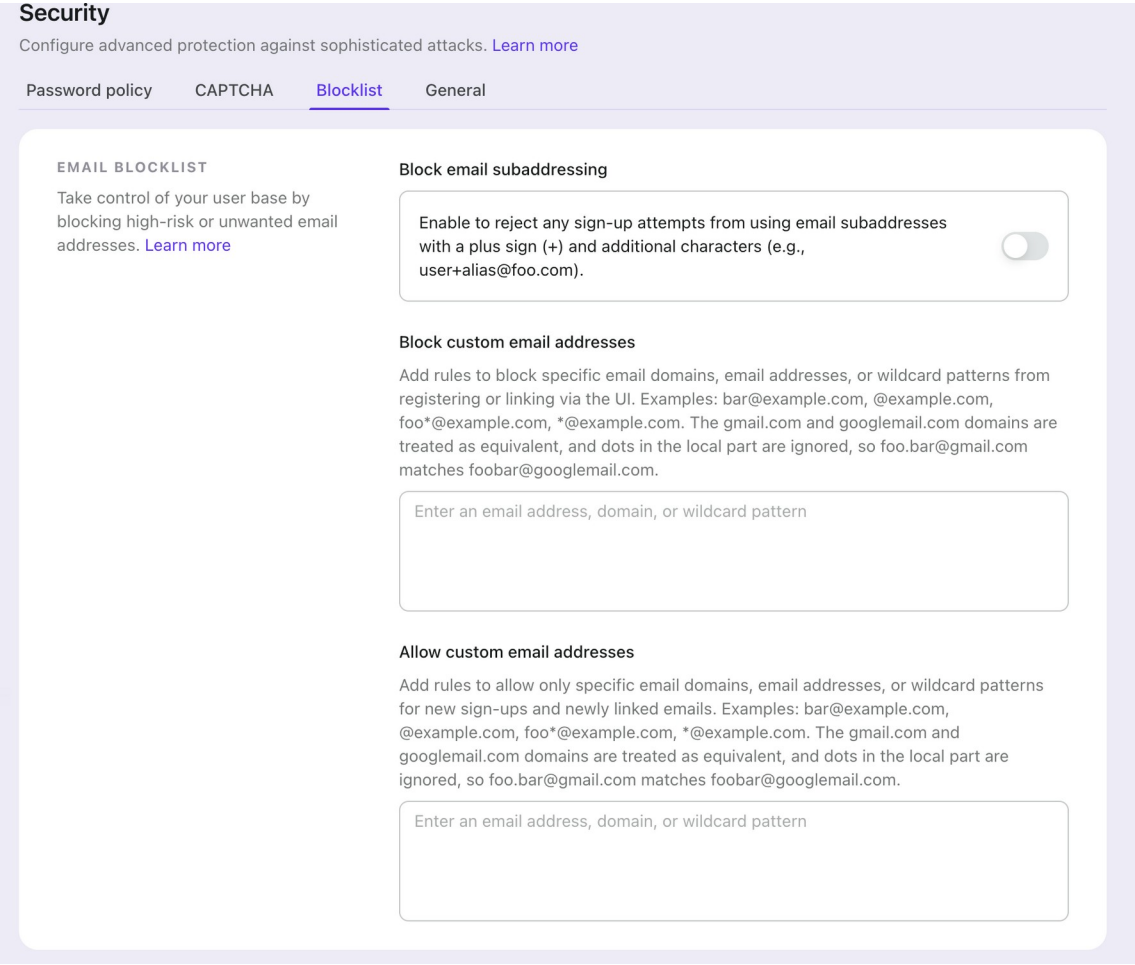
Staff use the bank’s identity provider



- Enterprise SSO connects to SAML and OIDC providers.
- Each connector is linked to an email domain.
- Staff do not need a second password for the platform.

Sign-up control

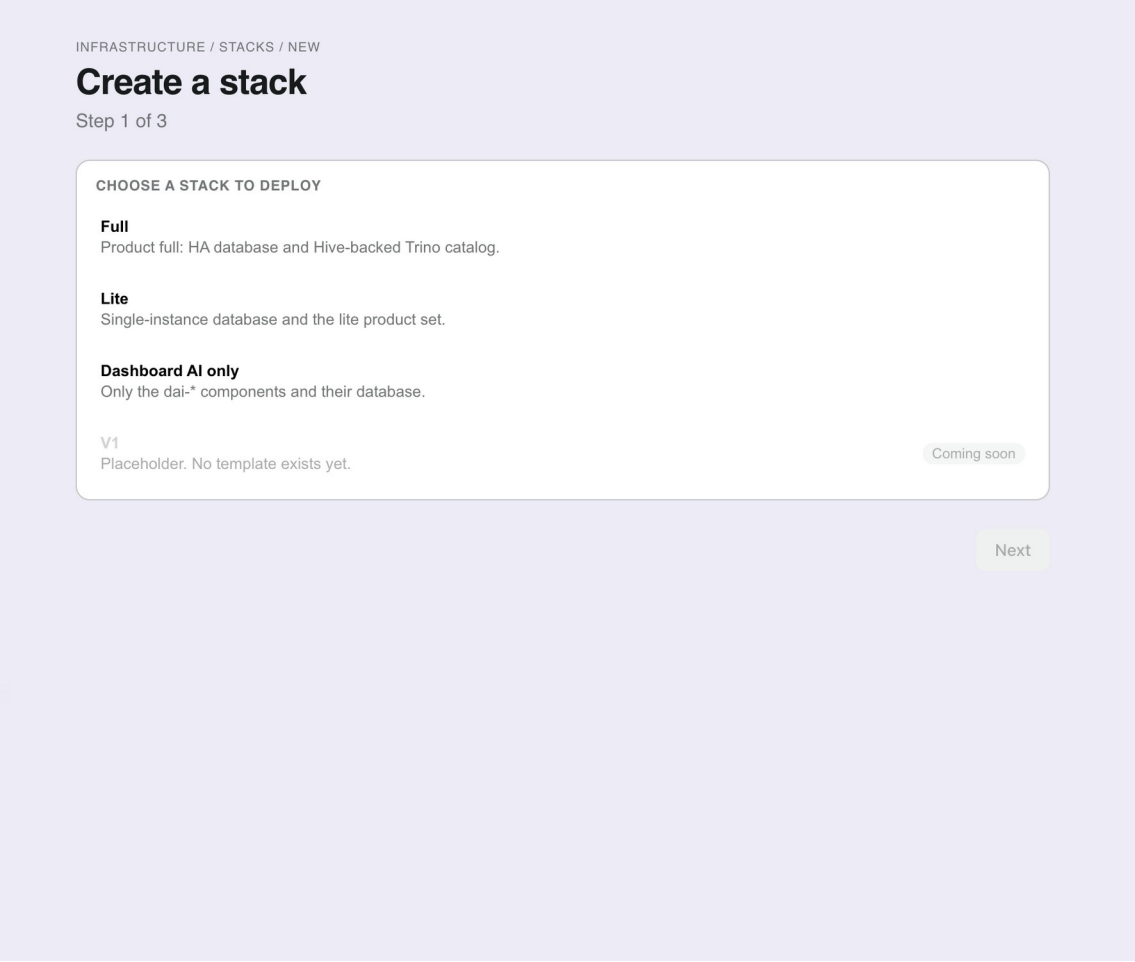
Block high-risk email domains at sign-up



- Reject disposable domains and subaddresses.
- Allow only the partner domains the bank approves.
- Rules apply to new sign-ups and new linked emails.

Stack wizard

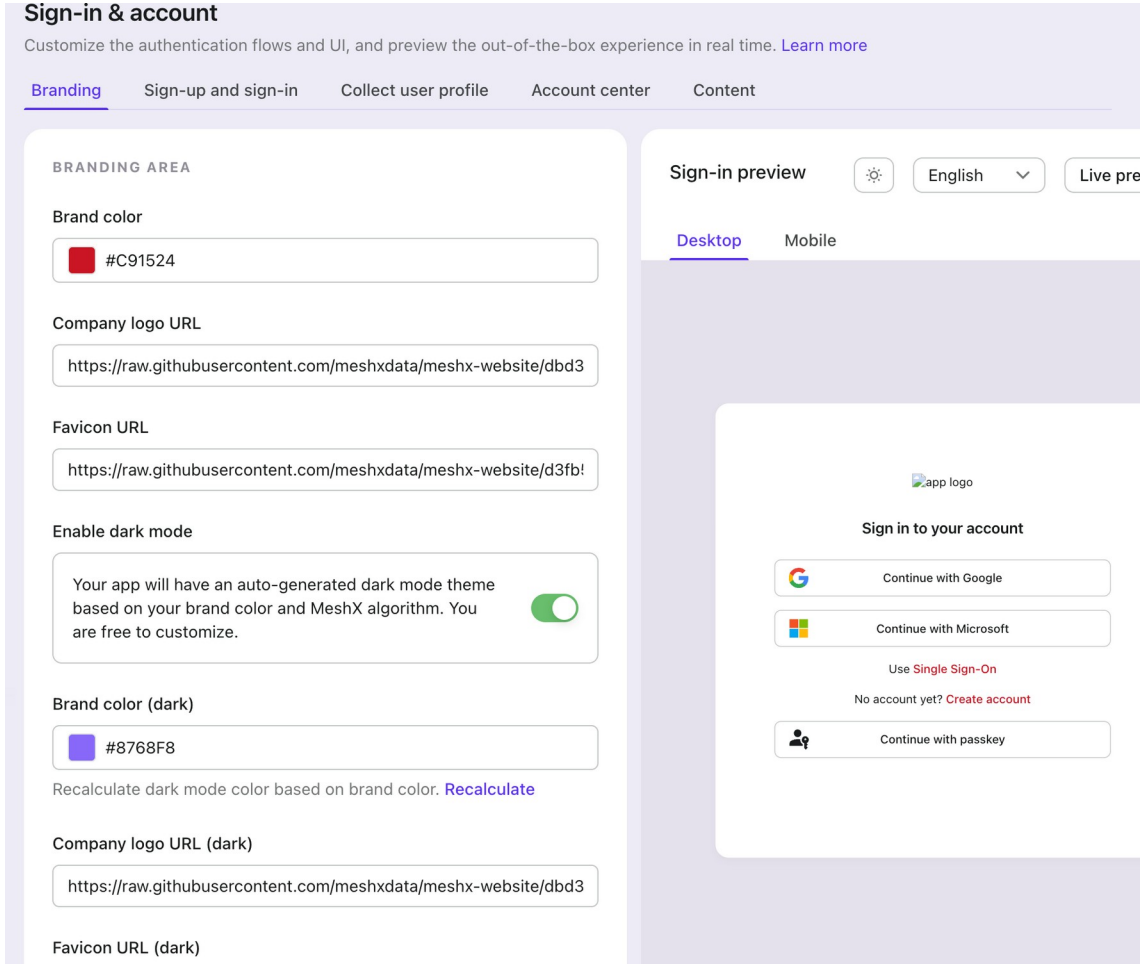
Create a stack from an approved template



- Pick Full, Lite, or Dashboard AI only.
- Set the tenant and mode, then review the YAML.
- Create. The stack appears in the stacks list.

Sign-in experience

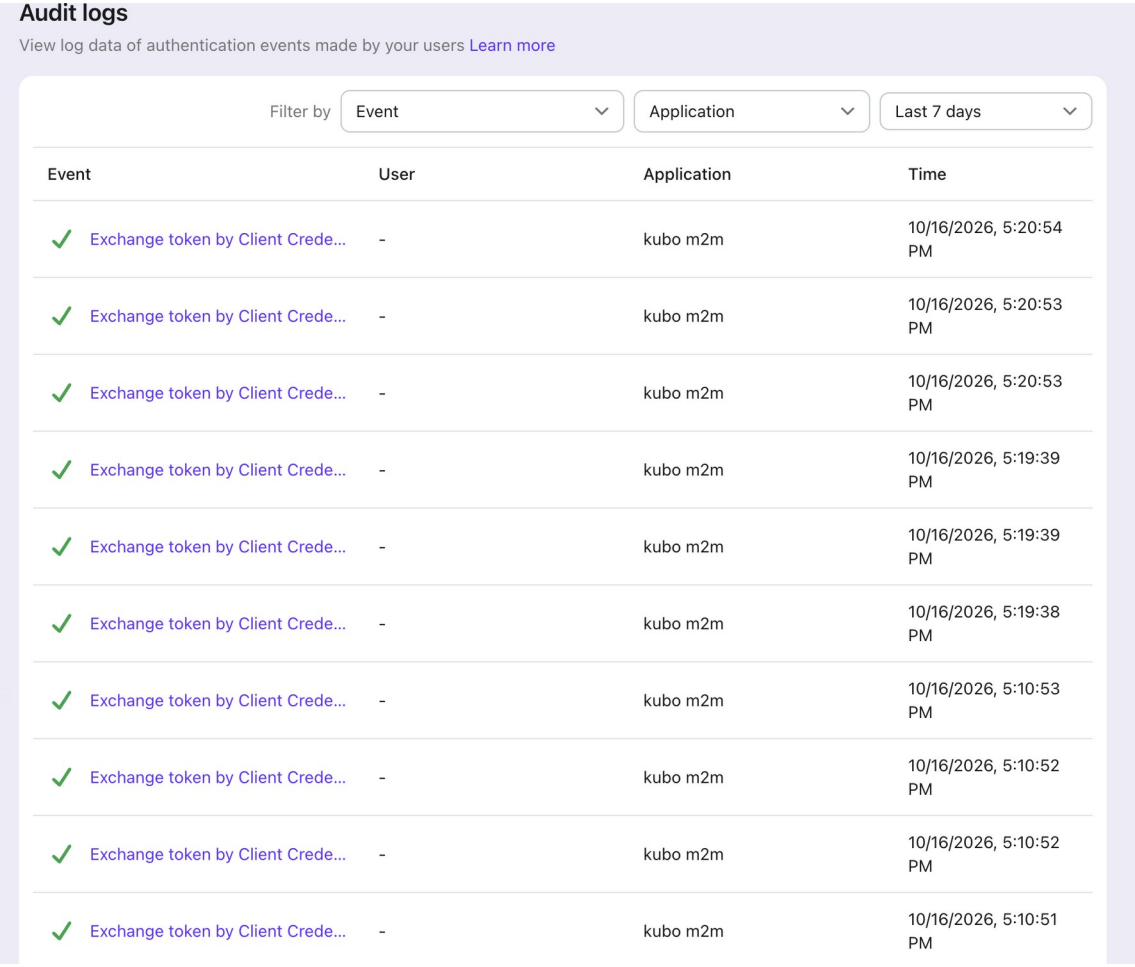
Branding and sign-in options for the tenant



- Set the logo, colors, and sign-in methods.
- Changes apply to the sign-in page at once.
- Custom sign-in UI is available without a license.

Audit trail

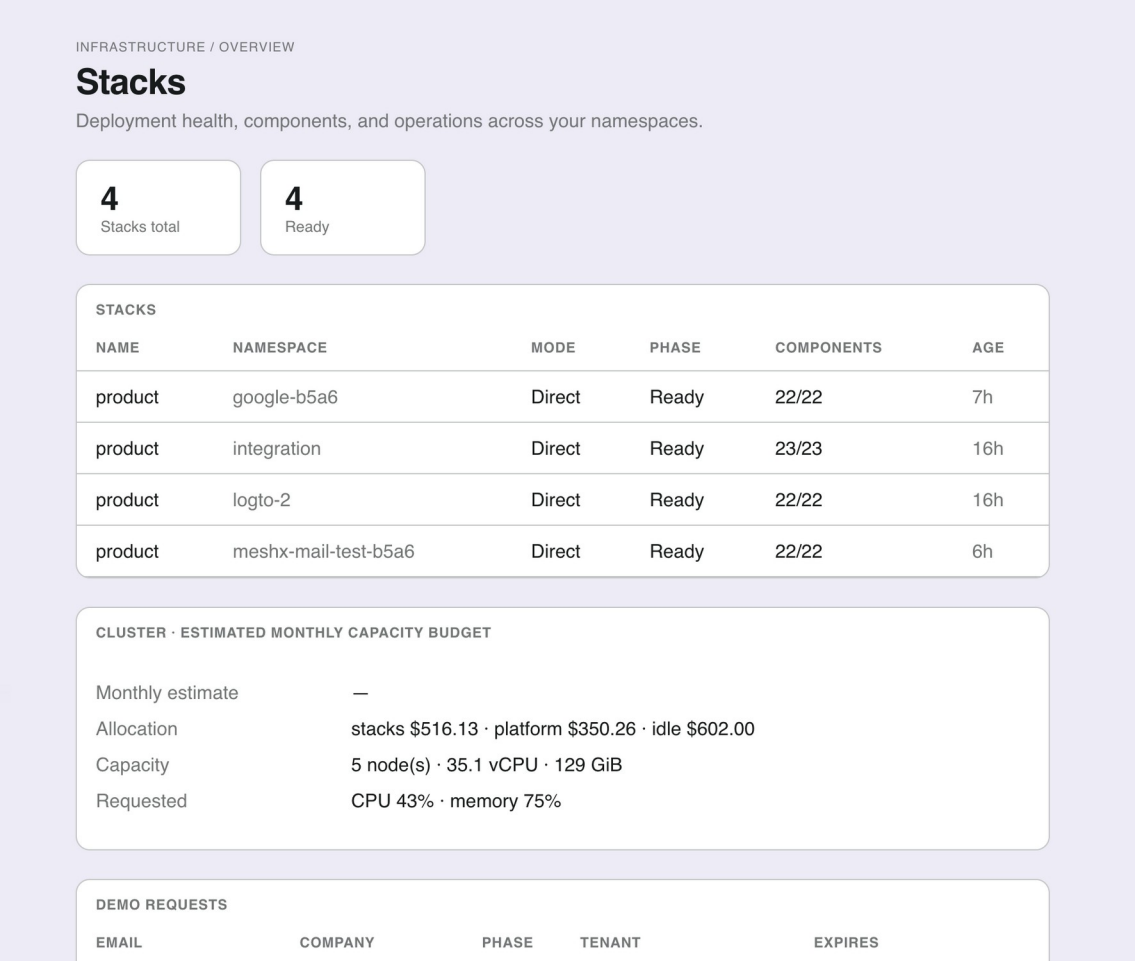
Each token exchange is recorded by application and time



- Filter by event, application, and time.
- Export the log for internal audit and regulator requests.
- Map each control to a log entry before the review.

Stacks

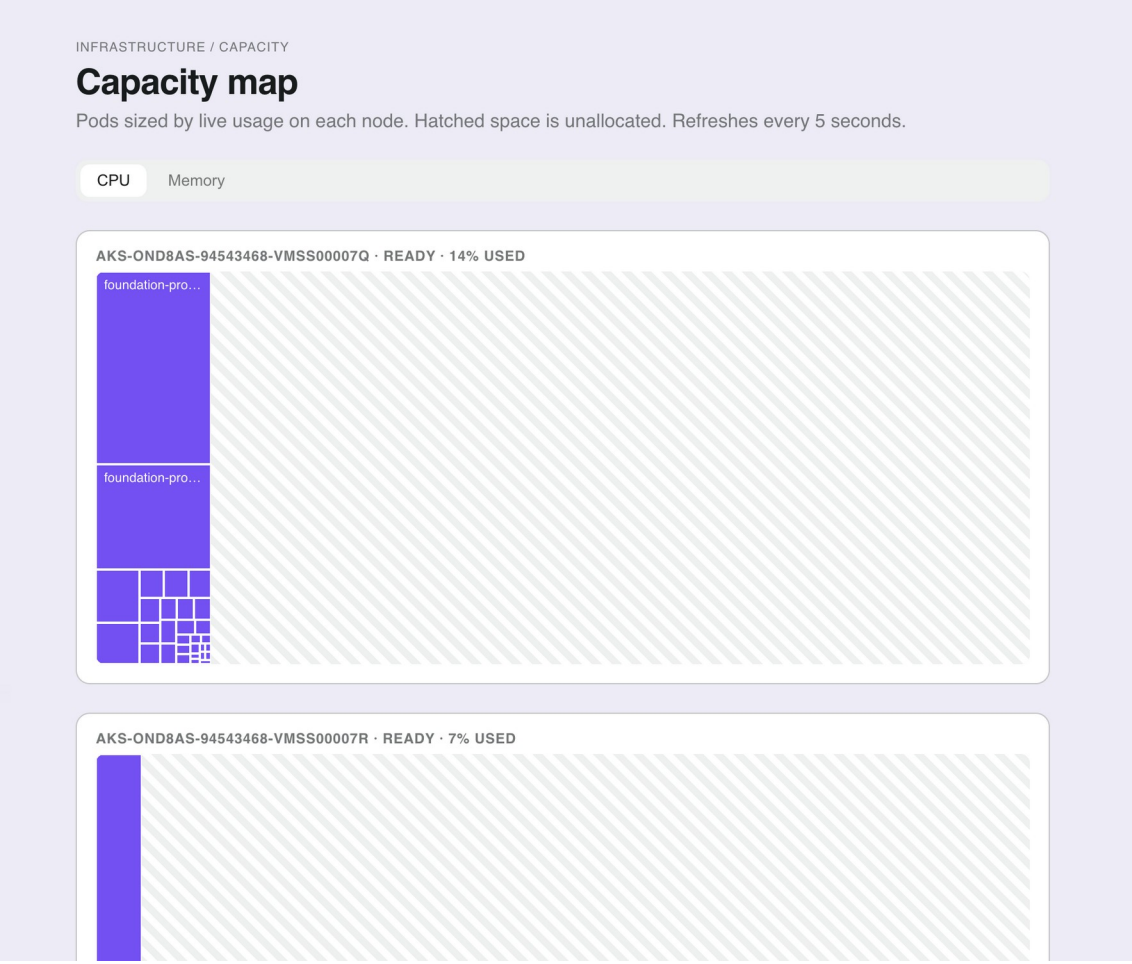
Phase and component status for each environment



- Each stack uses an approved template: Full, Lite, or Dashboard AI only.
- Component status, such as 22/22, shows if a deployment is complete.
- Confirm the core chart settings before a hardening claim is made.

Capacity map

Live CPU and memory use for each node and pod



- Pods are sized by live usage on each node.
- Hatched space shows capacity that is not allocated.
- Switch between CPU and memory.

How sign-in works

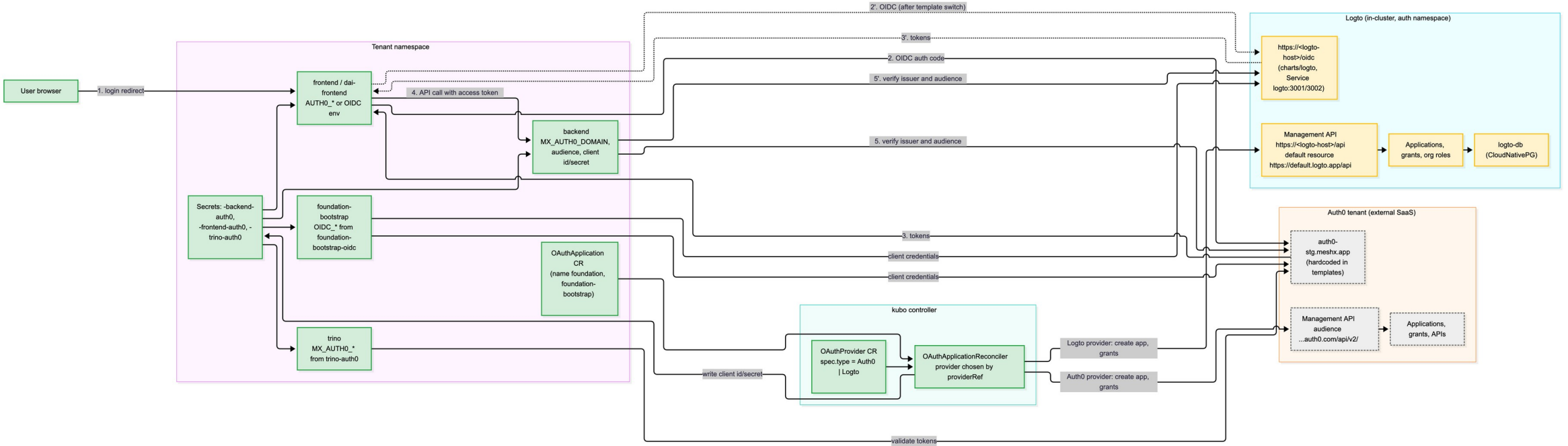
Each step, from the user to the identity provider

Step	From	To	What happens
1. Login redirect	User browser	Tenant frontend	The user starts sign-in.
2. OIDC authorization code	Frontend	Identity provider	The provider returns an authorization code.
3. Tokens	Identity provider	Frontend	The frontend receives the tokens.
4. API call with access token	Frontend	Backend	The backend receives the access token.
5. Verify issuer and audience	Backend	Identity provider	The backend checks each token before it is used.

Logto is the target provider. Product templates still point to Auth0 today.

Appendix: full sign-in diagram

Zoom in for the labels. Both sides of the flow are shown.



Secrets management

Secrets stay in a vault. They are not stored in charts or configuration files.

Component	State	Role
Vault (in-cluster, kubo-vault namespace)	Partial	Holds tenant secrets. Runs in the demo cluster.
External Secrets Operator	Partial	Copies secrets from the vault into tenant namespaces. About ten charts use it.
Store definitions per environment	Gap	Charts name the stores. Store definitions must be supplied per environment.
Plaintext secrets in configuration	Gap	Must be removed before a customer install. Verified check still to run.
Key rotation and access audit	Gap	Not built. Needed for the security review.

Secrets stay in the vault. Each environment supplies its own store definitions before any install.

Release and delivery in your network

Six steps, with the state of each

Step	State	Note
1. Build and sign the stack bundle	Partial	Bundles are supported. Signing is not yet verified.
2. Push the bundle to your registry	Designed	The cluster does not yet pull from your registry in a closed install.
3. Operator detects the new version	Proven	Pull-based reconcile in Direct or Flux mode.
4. Your approval gate	Gap	Not built.
5. Promotion to the target stack	Partial	Ordered apply and cleanup. No promotion across environments.
6. Rollback to a prior version	Gap	Not built as a user workflow.

No service level is quoted yet. The release interval is agreed with your change team.

Inference inside your boundary

Bedrock-only operation. Not yet verified in a running environment.

Requirement	State	Test to pass
Bedrock as the model provider	Gap	A chat request completes with no public call
Approved models and region	Gap	Only approved model IDs are reachable
IAM role for the service	Gap	Requests are signed with the role. No static key.
Private endpoint and DNS	Gap	The model name resolves to the private endpoint
Logs and keys	Gap	Invocation logs are kept in your account and encrypted with your key

A test environment is built first. The results are reported before the next review.

Decisions we need from you

Your answers set the order of the work

- Which AWS services are approved, and in which region?
- Does the platform run in AWS, on premises, or both?
- Do you need GPU nodes, or access to your existing model endpoint?
- Which models are approved, and how is access given?
- Who runs the platform day to day?
- Which operator access model do you require?
- Is identity federation needed from day one?
- Where does the data target run, and who owns the network path?

Next steps

Items not yet built. Owners and dates are to be confirmed.

Action	Owner	Timing
Prove Bedrock-only operation in a new test environment	meshX	Before the next review
Pull images from your registry, with a no-outside-access test	meshX platform	To confirm
Default to Logto for closed-network installs	meshX platform	To confirm
Add your approval gate and promotion steps to the release flow	To assign	To confirm
Build a user rollback workflow	To assign	To confirm
Plan GPU and on-premises model support after your answer	To assign	After your answer
Choose the data sharing network option with your network team	meshX with your network team	After your answer